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Risa 3D

Analysis of a Stable Beam

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Procedure:

Create a stable beam using Risa 3D, give the output of the beam and compare results to handwritten calculations.

Results:

The following table shows the results of my beam after applying load combinations, joint coordinates, boundary conditions, and solving via Risa 3D.

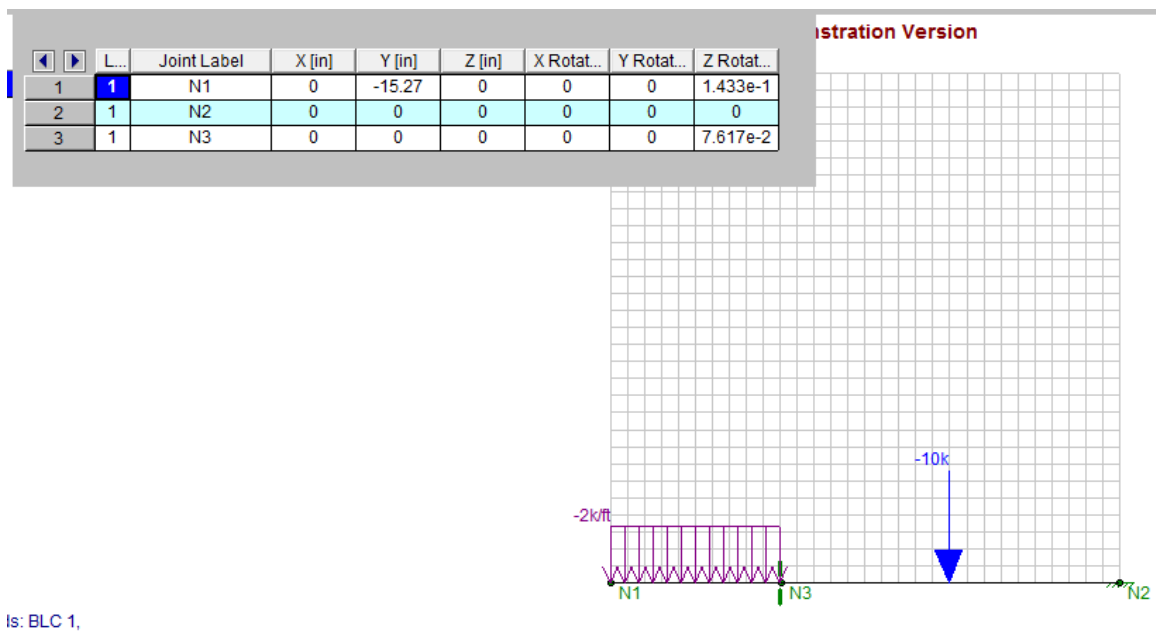
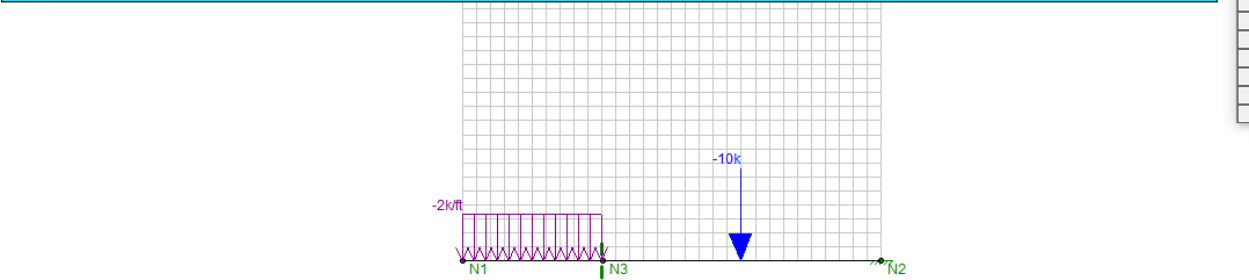


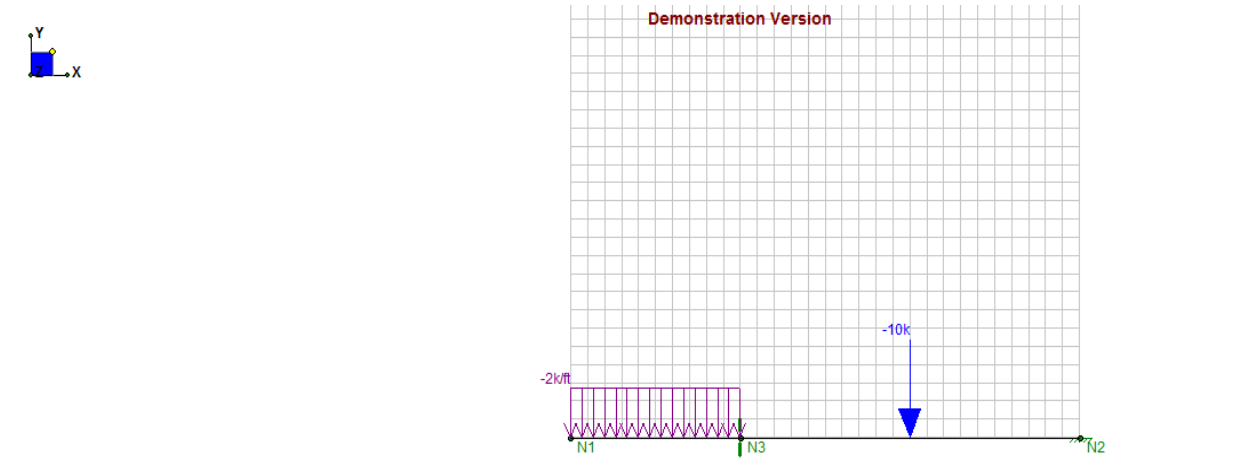
Table 1 : Beam results (Joint Deflections)

3D Maximum Member Section Forces (By Combination)															
Sections			Maximums			End Reactions									
	LC	Member Label		Axial[k]	Loc[ft]	y Shear[k]	Loc[ft]	z Shear[k]	Loc[ft]	Torque[k-ft]	Loc[ft]	y-y Moment[k-ft]	Loc[ft]	z-z Moment[k-ft]	Loc[ft]
1	1	M1	max	0	0	10.61	10.313	0	0	0	0	0	0	100	10
2			min	0	0	-20	10	0	0	0	0	0	0	-12.205	30



C 1,

Table 2 : Beam results (Member Forces)

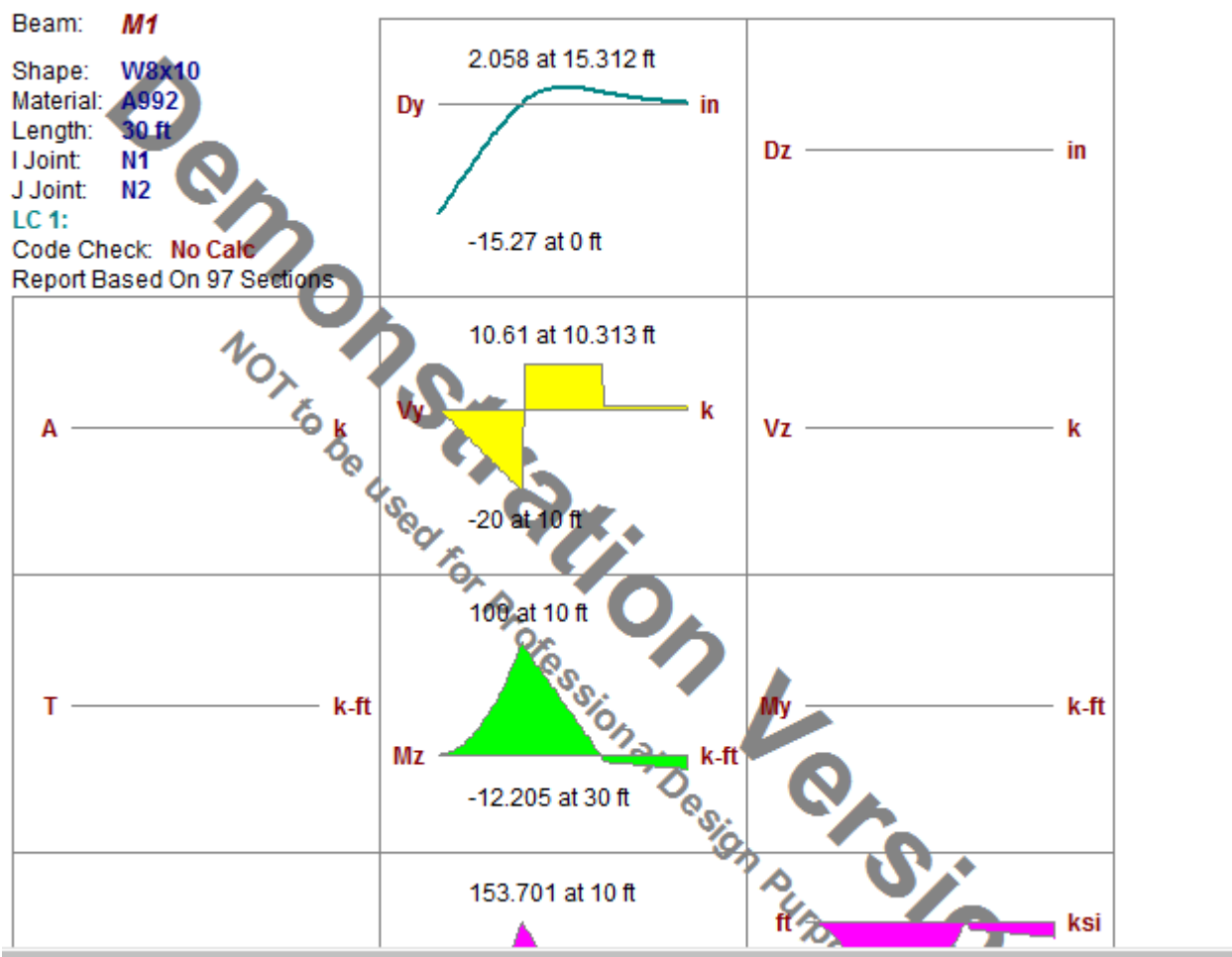


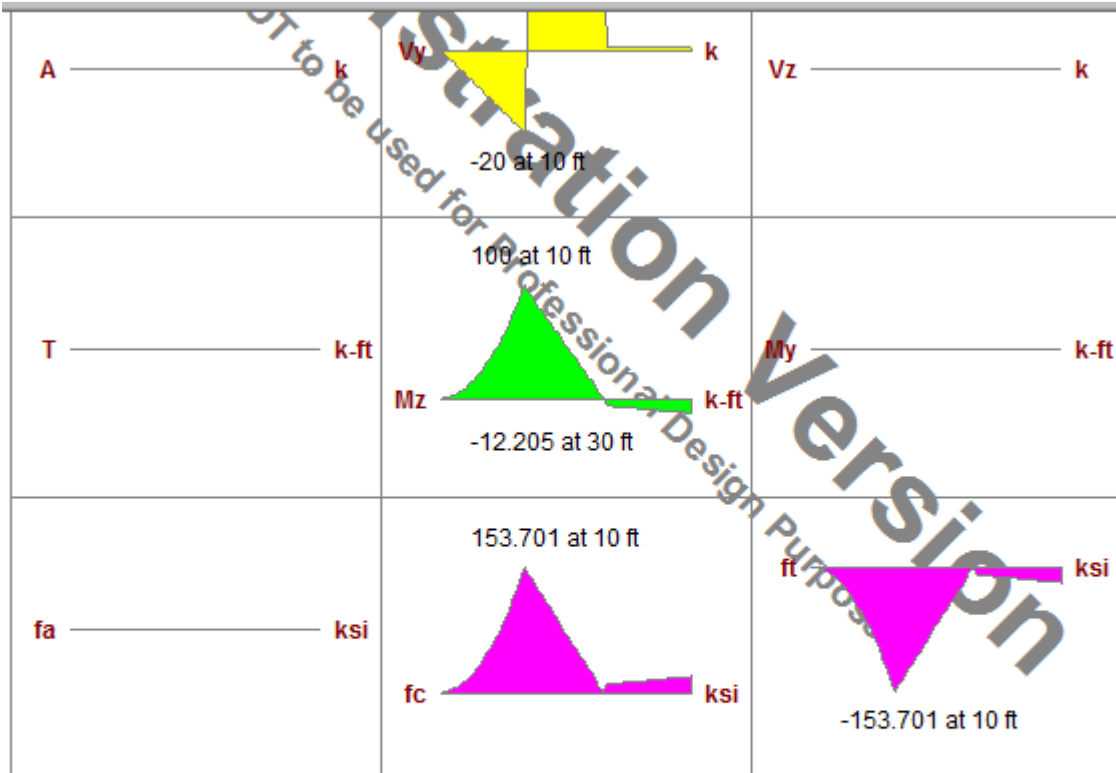
Loads: BLC 1,

Figure 1: Screenshot of Beam

Conclusion:

After analyzing the information given to me I was able to easily and efficiently navigate and create the model shown in my results section above. After analyzing my results from my Risa 3D data with my handwritten calculations I came across a discrepancy. I believe this may be due to the fact that I applied a hinge at B or there may have been a better method I could have taken to solve the beam to bring me to an equivalent analysis with RISA. I also believe that if the max allowable stress is 45 KSI this section would not be a good choice. I say this because after applying distributed loads and joint loads to the beam as shown, I arrived at a max of 153.701 ksi for the beam analysis. Below is a detailed view of the structural analysis given by Risa 3D.





AISC 15th(360-16): ASD Code Check

Direct Analysis Method

- P-Delta analysis required for all AISC 360-16 Load Combinations -

Max Defl Ratio **L/15**